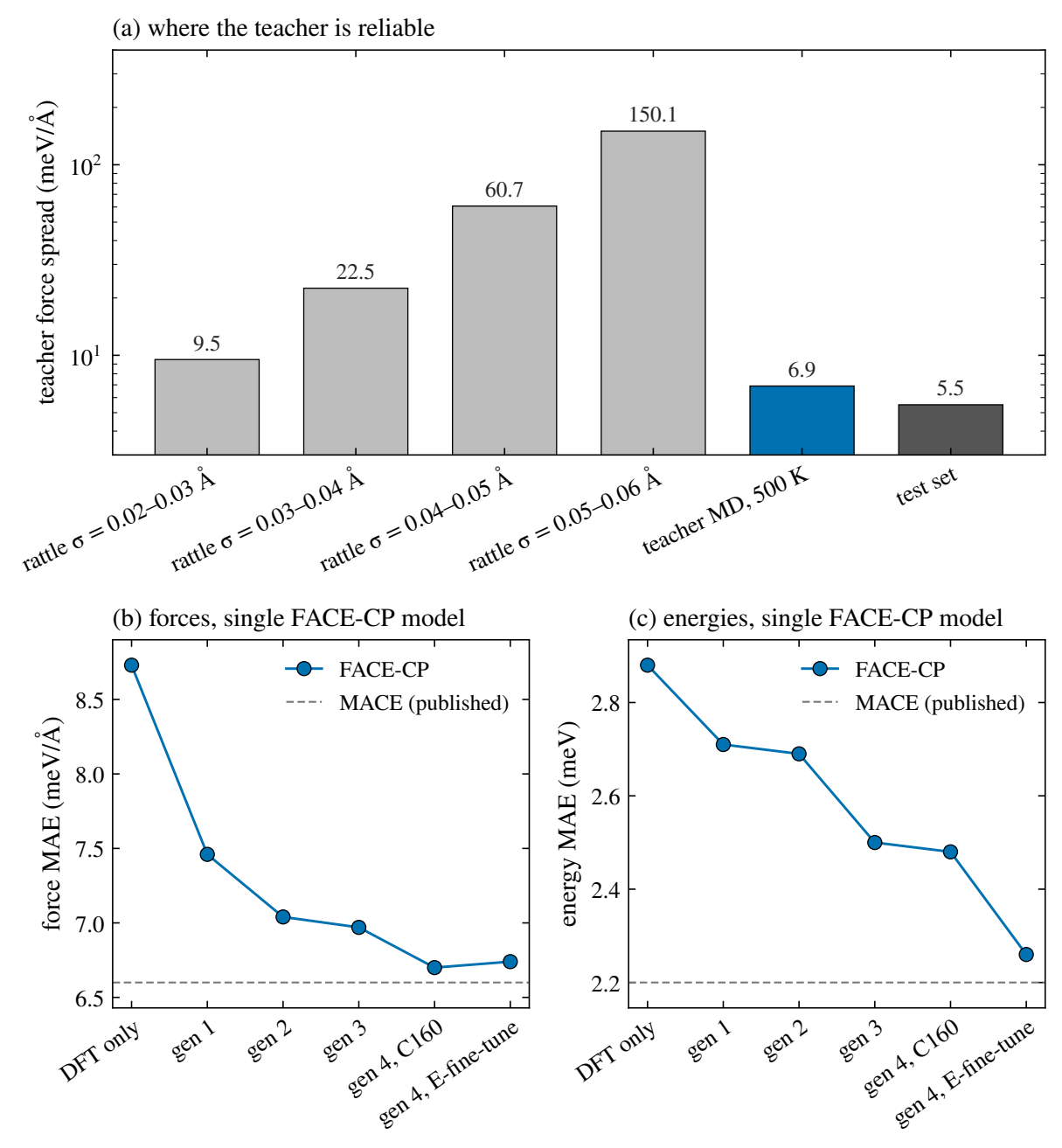
(a) where the teacher is reliable
teacher force spread (meV/Å)
10^2
10^1
9.5
22.5
60.7
150.1
6.9
5.5
rattle σ = 0.02–0.03 Å
rattle σ = 0.03–0.04 Å
rattle σ = 0.04–0.05 Å
rattle σ = 0.05–0.06 Å
teacher MD, 500 K
test set
(b) forces, single FACE-CP model
force MAE (meV/Å)
8.5
8.0
7.5
7.0
6.5
FACE-CP
MACE (published)
DFT only
gen 1
gen 2
gen 3
gen 4, C160
gen 4, E-fine-tune
(c) energies, single FACE-CP model
energy MAE (meV)
2.8
2.6
2.4
2.2
FACE-CP
MACE (published)
DFT only
gen 1
gen 2
gen 3
gen 4, C160
gen 4, E-fine-tune